WORKED SOLUTION

STEP II 2001, Question 12

Probability and Statistics • Lottery

STEP II 2001, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q12

The key idea

The syndicate is guaranteed exactly one winning ticket, so everything depends on how many *other* punters picked the winning number. That count is binomial with a huge number of trials and a tiny success probability, which is exactly the situation where a Poisson distribution takes over. The expected share is then W times the expectation of $1/(1+X)$, a sum that can be done in closed form by absorbing the $k+1$ into the factorial.

Solution

Setting up

The syndicate bought N tickets, so the other punters bought $3.8N - N = 2.8N$ tickets. Let X be the number of these non-syndicate tickets carrying the winning number.

Each non-syndicate punter chooses a number uniformly at random, independently of the others and of the draw, so whatever the winning number is, each of their tickets matches it with probability $1/N$, independently. Hence

$$X \sim \mathrm{B}\left(2.8N,\ \tfrac{1}{N}\right).$$

Since N is very large, $2.8N$ is large and $1/N$ is small while the mean $2.8N \cdot \frac{1}{N} = 2.8$ stays fixed, so X is approximately Poisson with mean $\lambda = 2.8$:

$$\mathrm{P}(X = k) \approx \frac{\mathrm{e}^{-2.8}\,(2.8)^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

The syndicate holds exactly one ticket with each number, so it always holds exactly one winning ticket, and the total number of winning tickets is $T = 1 + X$.

The most probable number of winning tickets

Write $p_k = \mathrm{P}(X = k)$. Comparing neighbouring probabilities,

$$\frac{p_k}{p_{k-1}} = \frac{\lambda}{k} = \frac{2.8}{k},$$

which is greater than 1 for $k \leq 2$ and less than 1 for $k \geq 3$. So $p_0 < p_1 < p_2$ and $p_2 > p_3 > p_4 > \cdots$, and the most probable value of X is 2. (The same holds for the exact binomial: there $p_k/p_{k-1} = \frac{(2.8N-k+1)}{k(N-1)}$, which exceeds 1 exactly when $k < 2.8 + \frac{1}{N}$.)

Therefore the most probable number of winning tickets is

$$T = 1 + 2 = 3.$$

The expected net loss

The jackpot is shared equally among the $T = 1 + X$ winning tickets, and the syndicate holds one of them, so it receives $\dfrac{W}{1+X}$. It paid £N, so its net loss is

$$L = N - \frac{W}{1+X}, \qquad \text{and} \qquad \mathrm{E}(L) = N - W\,\mathrm{E}\left(\frac{1}{1+X}\right).$$

Ask yourself How do we sum $\sum \frac{1}{k+1} \cdot \frac{\mathrm{e}^{-\lambda}\lambda^k}{k!}$?

The trick is that $(k+1)\cdot k! = (k+1)!$, so the awkward factor $\frac{1}{k+1}$ merges into the factorial. Multiplying and dividing by λ turns the sum into a Poisson series with one term missing:

$$\begin{aligned}\mathrm{E}\left(\frac{1}{1+X}\right) &\approx \sum_{k=0}^{\infty} \frac{1}{k+1} \cdot \frac{\mathrm{e}^{-\lambda}\lambda^k}{k!} = \frac{\mathrm{e}^{-\lambda}}{\lambda} \sum_{k=0}^{\infty} \frac{\lambda^{k+1}}{(k+1)!} \\ &= \frac{\mathrm{e}^{-\lambda}}{\lambda} \sum_{j=1}^{\infty} \frac{\lambda^j}{j!} = \frac{\mathrm{e}^{-\lambda}}{\lambda}\left(\mathrm{e}^{\lambda} - 1\right) = \frac{1 - \mathrm{e}^{-\lambda}}{\lambda},\end{aligned}$$

where we put $j = k+1$ and used $\sum_{j\geq 0} \lambda^j/j! = \mathrm{e}^{\lambda}$, removing the $j = 0$ term, which is 1. With $\lambda = 2.8 = \frac{14}{5}$, this is $\frac{5}{14}\left(1 - \mathrm{e}^{-2.8}\right)$, and so

$$\mathrm{E}(L) \approx N - \frac{5\left(1 - \mathrm{e}^{-2.8}\right)}{14} W,$$

as required.

Checks

- The same trick works for the exact binomial $X \sim \mathrm{B}(n, p)$: $\mathrm{E}\left(\frac{1}{1+X}\right) = \dfrac{1 - (1-p)^{n+1}}{(n+1)p}$, which tends to $(1 - \mathrm{e}^{-\lambda})/\lambda$ as $n \to \infty$ with $np = \lambda$. With $N = 1000$ the exact value is 0.33536, against the Poisson value $\frac{5}{14}(1 - \mathrm{e}^{-2.8}) \approx 0.33542$.

- A simulation of 10^5 draws with $N = 200$ gave an average share of $0.335W$, and 3 winning tickets was the most frequent outcome.
- Limiting case: if nobody else played ($\lambda \to 0$), $(1 - \mathrm{e}^{-\lambda})/\lambda \to 1$ and the syndicate collects the whole jackpot, as it should.

Common slips

- Using $3.8N$ rather than $2.8N$ for the other punters: the syndicate's own tickets are not random, and contribute exactly one winner.
- Writing $\mathrm{E}\left(\frac{W}{1+X}\right) = \frac{W}{1+\mathrm{E}(X)} = \frac{W}{3.8}$. Expectation does not pass through a non-linear function; in fact $\frac{1}{3.8} \approx 0.263$ is noticeably smaller than 0.335 (Jensen's inequality says it must be no larger, since $1/(1+x)$ is convex).

Beyond STEP

The approximation of $\mathrm{B}(n, \lambda/n)$ by $\mathrm{Po}(\lambda)$ is the "law of rare events", and first-year courses prove it by showing the probabilities, or the probability generating functions, converge. The calculation of $\mathrm{E}\big(1/(1+X)\big)$ is really an integral of the generating function, $\int_0^1 G_X(s)\,\mathrm{d}s$, a neat use of generating functions beyond computing means and variances. Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

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